

Ahmed Abdullah

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EDUCATION

- **National University of Computer and Emerging Sciences** Lahore, Pakistan
Bachelor of Science in Data Science; GPA 3.6
Aug. 2022 – May. 2026
 - **Honors:** Dean's List, Top of Batch
 - **Experience:** Teaching Assistant for OOP (CS1004) and COAL (EE2003)
 - **Relevant Courses:** Data Structures, Adv. Statistics, Artificial Intelligence, Operating Systems, Data Mining, Parallel and Distributed Computing & Data Warehousing.

EXPERIENCE

- **College of Engineering and Computing - Missouri S&T University** Missouri, US (Remote)
Research Assistant Feb 2025 – Present
 - **Computational Biology Research:** Led a team of 3 junior engineers to build a predictive model for tumor reduction using machine learning. Leveraged large-scale patient data from 50,000+ patients to assess the efficacy of cancer drugs over time. Contributed to designing the modeling pipeline and ensuring accurate insights for cancer growth.
- **Multimedia Mining & Search Group - Johannes Kepler University** Linz, AT (Remote)
Research Assistant Aug 2024 – Feb 2025
 - **Face-Voice Association with Missing Modalities:** Developed a deep learning system for associating faces with voices, boosting model accuracy by 12% on a widely used benchmark (VoxCeleb). Leveraged advanced audio-visual models like WavLM, ECAPA, and VGGVox to achieve state-of-the-art performance. Redesigned the training pipeline, introducing reusable modules that cut processing time by 10% and reduced setup time by 70%, significantly improving development efficiency.
- **Machine Intelligence Group (MInG) - FAST-NUCES** Lahore, PK (Hybrid)
Research Assistant Aug 2024 – Nov 2024
 - **Predicting Outcome for Ischemic Stroke:** Designed an AI system to predict stroke outcomes using MRI scans. Led the effort to improve model performance by enhance image data and increasing accuracy from 44% to 57%. Contributed to optimizing the image augmentation process, helping to integrate different data sources and set a new benchmark for stroke prediction.
- **UrduX Research Lab - FAST-NUCES** Islamabad, PK (Remote)
Research Intern Jun 2024 – July 2024
 - **Generative AI Research:** Contributed to a thesis project focused on identifying and reducing gender and race biases in AI-generated images. Analyzed over 10,000 images and played a key role in the implementation of strategies to mitigate biases, helping to reduce gender bias by 15% and racial bias by 12%.

PROJECTS

- **PolyHope-M – Hope Speech Classification:** Built a multilingual classification model using LLaMA and PyTorch to classify supportive speech online. Achieved 1st place out of 400+ submissions on CodaLab with top-ranked accuracy.
- **DermaScan3D – Skin Cancer Detection using CNNs:** Built a multimodal TensorFlow model with ResNet for skin cancer classification. Ranked in the top 1% (8000+ participants) on Kaggle.
- **VisionFind – Multimodal Search Engine using CLIP:** Built a semantic search engine using CLIP, HuggingFace, and PyTorch to retrieve images and text based on visual and textual queries.
- **CrickAI – LBW Review System using YOLO:** Built a video review tool for LBW decisions in cricket using YOLO for object detection. Utilized Roboflow for dataset preparation and Labelling for annotation.

PROGRAMMING SKILLS

- **Languages:** Python, C++, C, JavaScript.
- **Technologies:** PyTorch, TensorFlow, Keras, HuggingFace, MONAI, Scikit-learn, MLLib, Transformers, OpenCV.
- **Tools & Platforms:** LangChain, Flask, Firebase, Github, PowerBI, Tableau, Hadoop, PySpark, PostgreSQL, MongoDB.